02 Esophagus and stomach

Team: _____

Basic function (this is all you need): The esophagus moves the bolus to the stomach. The stomach churns it with acid and enzymes into chyme.

STRUCTURES TO DRAW AND LABEL

- Esophagus: about 25 cm, posterior to the trachea, passes through the esophageal hiatus of the diaphragm at T10
- Upper esophageal sphincter (skeletal) and lower esophageal sphincter (a thickening of smooth muscle, not a true anatomical sphincter)
- Stomach regions: cardia, fundus, body, pyloric part (pyloric antrum and pyloric canal), pylorus, pyloric sphincter
- Lesser curvature with the lesser omentum; greater curvature with the greater omentum
- Rugae (folds of mucosa and submucosa when the stomach is empty)
- Gastric pits leading into gastric glands

NERVES AND BLOOD SUPPLY

- Parasympathetic: vagus nerve (CN X) as the anterior and posterior vagal trunks
- Sympathetic: greater splanchnic nerves (T5 to T9) to the celiac ganglion
- Upper esophagus skeletal muscle: somatic fibers in CN X (recurrent laryngeal branches)
- Enteric plexuses: submucosal (in the submucosa) and myenteric (between the circular and longitudinal muscle). Present along the whole tube.
- Artery: Celiac trunk: left gastric, right gastric, right and left gastro-omental arteries, short gastric arteries

THE WALL UNDER THE MICROSCOPE

- Esophagus: nonkeratinized stratified squamous epithelium
- Esophageal glands (mucous) in the submucosa
- Esophageal muscularis externa: skeletal muscle in the upper third, mixed in the middle third, smooth in the lower third
- Esophagus has an adventitia, except the short abdominal part, which has a serosa
- Stomach: simple columnar epithelium made entirely of surface mucous cells; no goblet cells
- Stomach muscularis externa has three layers: inner oblique, middle circular, outer longitudinal
- Cardiac and pyloric glands are mostly mucous. Fundic (gastric) glands in the fundus and body have the full set of cells
- Serosa covers the stomach

CELLS AND WHAT THEY MAKE

Surface mucous cells	Thick alkaline mucus that protects the lining
Mucous neck cells	Thinner mucus
Parietal (oxyntic) cells	Hydrochloric acid and intrinsic factor
Chief (zymogenic) cells	Pepsinogen and gastric lipase
G cells (enteroendocrine, mostly in the pyloric antrum)	Gastrin
Other enteroendocrine cells	Local hormones such as somatostatin and histamine
Stem cells (isthmus and neck)	Replace the lining

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Analogy: a three-angle washing machine

The stomach is the only part of the tract with three muscle layers (oblique, circular, longitudinal) because it has to churn, not just push. Everywhere else has two.

Mnemonic: Parietal Pumps, Chief Chews

Parietal cells pump acid (and make intrinsic factor). Chief cells make pepsinogen, the protein chewer. G cells say Go (gastrin). Esophagus muscle: skeletal at the top, smooth at the bottom, mixed in the middle, the same order as swallowing.